F26-003 FYP-1 Mid-Evaluation

Investify

Educational stock research with evidence and checked explanations

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Problem

- News, price indicators and social discussion can disagree.
- A weak signal needs confidence and a record of its inputs.
- Historical leakage can inflate results before the user sees them.
- Fluent explanations can invent a number or a market cause.

The learner needs evidence they can inspect.

Panel feedback and actions

Feedback	Response	Status
PF-01	C1 harness, C2 verifier, C3 planned divergence	Partial
PF-02/03	Defined layer, eleven features, worked example	Partial
PF-04/05	Refined problem and pinned inventory	Closed
PF-06	Definitions, targets and observed results	Partial
PF-07	Profile, corruption and checked recovery	Closed
PF-08	Onboarding, Observe and per-step disclosures	Partial
PF-09	Mechanisms and design implications	Closed

Status describes the technical response; remaining evaluation stays in the plan.

Contributions C1–C3

- **C1: Timing harness and permanent canary.** Implemented; fault and boundary tests. Gate A remains partial.
- **C2: Deterministic verifier and checked fallback.** Implemented for template grammar; independent prose/JSON tests.
- **C3: News/social divergence ablation.** Planned; protocol fixed before the comparison.

No novelty claim in sentiment, XGBoost, SHAP-to-text or QLoRA. No profit claim.

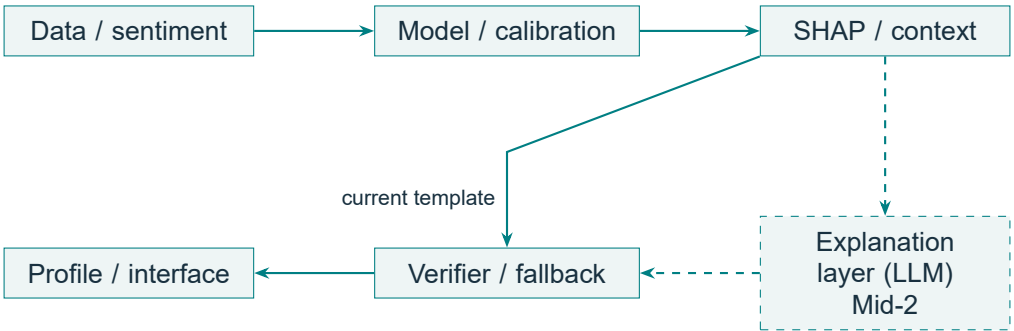
Market background and design decisions

Market mechanism	Design implication
Sessions and timestamps	Exchange calendar; backward joins
Earnings, macro and flows	Eleven features; no causal valuation model
Absorption and noise	Baselines and null-result reporting
Splits and changing regimes	Pin price basis and fit/evaluation periods
News tone versus model effect	Definitions beside SHAP effects

Model inventory and status

Component	Task	Status
ProsusAI FinBERT	Financial-headline sentiment	Executed, pinned original
CryptoBERT / Cardiff	Social sentiment comparison	Same frozen 24 posts
Project XGBoost	Eleven features to classes	Trained and evaluated
OpenAI GPT-6 teacher	Controlled prose examples	Six recorded; JSON assembled
Phi-3-mini	Local prompted explainer	Planned: Mid-2
Mistral	Explainer fallback	Planned, unexecuted
QLoRA	Student adaptation	Iteration 4

Architecture with the planned LLM layer



LLM prose and claims must preserve the trusted record and risk-policy outcome.

Requirements and metric targets

Observable acceptance

- All eleven features inspectable
- Complete profile; fixed probabilities
- Wrong prose/JSON rejected
- Observe headline and policy fallback
- Date, horizon, confidence and limits

Planned explanation targets

Numeric fidelity	≥ 0.98
Direction consistency	≥ 0.95
Claim recall	≥ 0.80
Unverifiable rate	≤ 0.15

Prediction: macro-F1 lift, Brier/reliability. Social: F1/recall/kappa. Users: time/errors/completion.

Live demonstration

1. **Onboarding / Decision**: purpose, Observe headline and diagnostic probabilities
2. **Evidence**: MACD value, meaning and SHAP effect; news tone versus effect
3. **Risk profile**: cautious/higher-tolerance thresholds, both Observe
4. **Explanation**: accept, inject 1.00%, reject and check fallback

Frozen multiclass NVDA / 21 December 2023. Profile and checking run live; model outputs are saved.

Leakage experiment and linear baseline

Five-session CV	Macro-F1	SD / lift
Clean XGBoost	0.2843	SD 0.0537
Linear	0.2796	SD 0.0559
Stratified random	0.3287	XGBoost lift -0.0444
Centred Bollinger fault	0.5390	Lift +0.2104

Final 186 cases, isotonic: XGBoost 0.2238; linear frozen 0.1800 / refit 0.1896.

Canary fires; clean prediction has no established useful advantage.

Calibration and abstention

Frozen mode	Buy / Hold / Sell	Brier
Raw	40 / 51 / 95	0.7929
Isotonic	186 / 0 / 0	0.6105
Temperature	40 / 51 / 95	0.6667

T = 1000: confidence approaches one-third.

- Brier $2/3$ is the constant one-third forecast.
- Best profile threshold is 50%; real confidence is about 33%.
- Both profile examples stay Observe. Controlled 0.70 fixtures show the gate behavior.

Illustrative explanation and claim checks

Illustrative design example, not a system output: Hold 58%; news +0.31 and social -0.18 disagree; divergence SHAP +0.14 supports Hold; RSI SHAP -0.06 opposes Hold.

Claim	Planned check
Hold at 58%	Record class and probability
News/social values	Recorded feature values and coverage
Divergence supports Hold	Selected-class SHAP sign
RSI opposes Hold	Does not establish support for Buy
Volatility caused the move	Refuse unsupported causal claim

Current checker: template grammar only. Free-prose claim binding: Mid-2.

Testing and CI/CD

Observed local tests

Portable	183
CI guards	7
Legacy UI	24
Presenter	19
Study selectors	4

Hosted delivery

- 2 jobs $\times$ 2 OS = 4 checks
- 1 gated delivery job
- Same-commit source/PDF checks
- Downloadable artifact; hosting later

Counts are distinct scopes, not a synthetic total. Actual commit/run: report Appendix D.

Remaining plan and Q&A

Stage	Remaining work
Final-1 / Iteration 2	Broader data, divergence ablation, engine/service and deployment
Mid-2 / Iteration 3	Prompted explainer, constrained claims, NFR/parity and acceptance preparation
Final-2 / Iteration 4	QLoRA comparison, broader evaluation and participant/external-install evidence

Implemented: template + verifier. Predictor usefulness and user benefit remain evaluation questions. **Questions**